Dimensional Concordance of State-Balance Formulations and Geometric Curvature

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1 Dimensional Consistency in Non-Canonical Metric-Substrate Formulations

Theoretical physics relies on rigorous dimensional homogeneity: regardless of the ontological interpretation assigned to physical entities, every term in an equation of motion must balance across fundamental base dimensions $[M]$, $[L]$, and $[T]$. When theoretical physicists encounter non-canonical paradigms—such as treating spacetime curvature as the thermodynamic relaxation or dynamic strain of an underlying discrete causal substrate—the concepts often appear unfamiliar or counterintuitive.

A primary concern is whether recasting physical interactions in terms of state-balance discrepancies, substrate debt, and causal relaxation introduces dimensional anomalies into general relativistic relationships. As confirmed by the mathematical architecture of the framework (S1), modifying active numerical code across simulation nodes is neither necessary nor advisable; the existing computational infrastructure already executes valid physical transformations. Instead, the task is strictly theoretical: providing an exact analytical mapping accompanied by comprehensive dimensional balance tables. This exposition demonstrates that the state-balance formulation is fully homeomorphic to canonical force, energy, and geometric curvature, showing that the unfamiliar vocabulary creates zero dimensional contradiction with canonical physics.

2 Equivalence of State-Balance Gradients to Canonical Force and Energy

In canonical mechanics and field theory, energy E possesses dimensions of $[\mathrm{ML}^2\mathrm{T}^{-2}]$, while force is defined either dynamically as the time-derivative of linear momentum $F = \mathrm{d}p/\mathrm{d}t$ or spatially through the negative gradient of a scalar potential $F = -ablaV$, yielding dimensions of $[\mathrm{MLT}^{-2}]$.

The state-balance substrate paradigm introduces the balance discrepancy level $\Delta\mathcal{B}$ as the primary scalar measure of local configurational departure from equilibrium. Rather than postulating a phenomenological energy term out of whole cloth, the theory establishes an exact dimensional identity:

$$\Delta E \equiv \Delta\mathcal{B}, \quad [\Delta\mathcal{B}] = \mathrm{ML}^2\mathrm{T}^{-2}$$

When a spatial gradient exists across adjacent substrate elements, the localized asymmetry in state balance induces a directed operational persistence—the effective drive toward equilibrium. The spatial gradient of this discrepancy constitutes the executive force:

$$\mathcal{F}_\mu = -\partial_\mu(\Delta\mathcal{B})$$

Evaluating the dimensional structure of this gradient yields:

$$[\mathcal{F}_\mu] = [\partial_\mu] \cdot [\Delta\mathcal{B}] = \mathrm{L}^{-1} \cdot \left(\mathrm{ML}^2\mathrm{T}^{-2}\right) = \mathrm{MLT}^{-2}$$

This corresponds exactly to the canonical unit of force (the Newton, $\mathrm{N} = \mathrm{kg} \cdot \mathrm{m} \cdot \mathrm{s}^{-2}$). When integrated over a localized boundary surface $\partial\Omega$, this spatial rate of change produces the canonical stress tensor Π_{ij} and boundary momentum flux. The unfamiliar terminology of "state-balance discrepancy" thus conceals no dimensionally exotic physics: it operates as a standard conservative potential in the spatial manifold, driving boundary traction vectors through identical mathematical and dimensional mechanics (S1).

3 Comprehensive Dimensional Alignment Matrix

To provide complete clarity and eliminate any lingering ambiguity regarding dimensional equivalence, the fundamental quantities of canonical physics and their state-balance substrate counterparts are systematically mapped below.

Canonical Physical Quantity	Canonical Dimension	SI Unit	Substrate State-Balance Counterpart	Mathematical Expression	Substrate Dimension
Scalar Energy (E)	$\mathrm{ML^2T^{-2}}$	J ($\mathrm{kg \cdot m^2/s^2}$)	Balance Discrepancy Level ($\Delta\mathcal{B}$)	$\Delta E = \Delta\mathcal{B}$	$\mathrm{ML^2T^{-2}}$
Translational Force (F)	$\mathrm{MLT^{-2}}$	N ($\mathrm{kg \cdot m/s^2}$)	Balance Gradient / Executive Persistence ($\mathcal{F}$)	$\mathcal{F}_\mu = -\partial_\mu(\Delta\mathcal{B})$	$\mathrm{MLT^{-2}}$
Boundary Stress / Pressure (Π)	$\mathrm{ML^{-1}T^{-2}}$	Pa ($\mathrm{N/m^2}$)	Integrated Boundary Surface Traction (Π_{ij})	$\Pi_{ij} = \frac{1}{A}\int \mathcal{F}_i \, \mathrm{d}A_j$	$\mathrm{ML^{-1}T^{-2}}$
Energy-Momentum Tensor ($T_{\mu u}$)	$\mathrm{ML^{-1}T^{-2}}$	$\mathrm{J/m^3}$ (Pa)	Bulk Substrate Balance Density ($T^{\mathrm{eff}}_{\mu u}$)	$T_{\mu u} = \partial_\mu\mathcal{B}\,\partial_u\tau$	$\mathrm{ML^{-1}T^{-2}}$
Causal Duration (t, τ)	T	s	Debt Dissipation / Relaxation Time (τ_{relax})	$\mathrm{d}t \sim \Gamma^{-1}_{\mathrm{evap}}\,\mathrm{d}\mathcal{D}$	T
Metric Tensor ($g_{\mu u}$)	1 (dimensionless)	—	Substrate Normalized Strain Configuration ($g_{\mu u}$)	$g_{\mu u} = \eta_{\mu u} + h_{\mu u}$	1
Ricci Curvature Tensor ($R_{\mu u}$)	$\mathrm{L^{-2}}$	$\mathrm{m^{-2}}$	Spacetime Curvature Proxy / Substrate Invariant	$R_{\mu u} = L^{-2}_{\mathrm{sub}} g_{\mu u}$	$\mathrm{L^{-2}}$

As the matrix reveals, every derived quantity in the substrate formulation maps bijectively into canonical dimensions, with zero leftover or anomalous dimensional units

across all tiers of the theory (S1, S3).

4 Bridge Between Dimensionless Metric and Curvature Tensors

A persistent source of confusion in unconventional geometric formulations is the apparent dimensional mismatch between the metric tensor and curvature tensors. In standard differential geometry, the Riemannian metric tensor is dimensionless:

$$[g_{\mu u}] = 1$$

In contrast, the Ricci curvature tensor $R_{\mu u}$ is constructed from Christoffel symbols and their spatial derivatives ($\partial\Gamma \sim \partial^2 g$), which naturally possess inverse-area dimensions:

$$[R_{\mu u}] = \mathrm{L}^{-2}$$

In conventional General Relativity, this relationship is satisfied through the Einstein field equations via the dimensional coupling constant κ:

$$G_{\mu u} = R_{\mu u} - \frac{1}{2} R g_{\mu u} = \frac{8\pi G}{c^4} T_{\mu u}$$

$$[\kappa] = \left[\frac{8\pi G}{c^4}\right] = \frac{\mathrm{M}^{-1}\mathrm{L}^{3}\mathrm{T}^{-2}}{\mathrm{L}^{4}\mathrm{T}^{-4}} = \mathrm{M}^{-1}\mathrm{L}^{-1}\mathrm{T}^{2}$$

When multiplied by the stress-energy tensor $[T_{\mu u}] = \mathrm{ML}^{-1}\mathrm{T}^{-2}$, the resulting product reproduces the exact curvature dimensions:

$$[\kappa \cdot T_{\mu u}] = (\mathrm{M}^{-1}\mathrm{L}^{-1}\mathrm{T}^{2})(\mathrm{ML}^{-1}\mathrm{T}^{-2}) = \mathrm{L}^{-2} = [R_{\mu u}]$$

In the substrate paradigm, an analogous dynamic identity connects the relaxed metric directly to the asymptotic Ricci tensor:

$$R_{\mu u} = L_{\mathrm{sub}}^{-2} g_{\mu u}$$

Here, the bridging parameter L_{sub} is not an unconstrained fitting constant, but the fundamental coherence length scale of the vacuum substrate (S2):

$$L_{\mathrm{sub}} \equiv \frac{c^2}{a_0}$$

where c is the speed of light ($[\mathrm{LT}^{-1}]$) and a_0 is the critical acceleration scale ($[\mathrm{LT}^{-2}]$). Evaluating the dimensions of L_{sub}:

$$[L_{\mathrm{sub}}] = \frac{\mathrm{L}^2\mathrm{T}^{-2}}{\mathrm{LT}^{-2}} = \mathrm{L} \implies \left[L_{\mathrm{sub}}^{-2}\right] = \mathrm{L}^{-2}$$

Multiplying this scale factor by the dimensionless metric $g_{\mu u}$ yields:

$$[R_{\mu u}] = [L_{\mathrm{sub}}^{-2}] \cdot [g_{\mu u}] = \mathrm{L}^{-2} \cdot 1 = \mathrm{L}^{-2}$$

The dimensional identity is verified with complete mathematical consistency. Dimensional continuity between the dimensionless metric and the curvature tensor is preserved without ambiguity.

5 Synthesis and Theoretical Consistency

The foregoing mathematical and dimensional exposition demonstrates that the state-balance formulation and its substrate dynamics introduce no dimensional discrepancies into the foundations of theoretical physics. By directly establishing that:

1. The state-balance discrepancy $\Delta\mathcal{B}$ is identical in dimension to canonical scalar energy ($\mathrm{ML^2T^{-2}}$);
2. The directional gradient $-\partial_\mu(\Delta\mathcal{B})$ reproduces the exact physical dimensions of classical force ($\mathrm{MLT^{-2}}$); and
3. The coupling between the dimensionless metric $g_{\mu u}$ and the Ricci curvature tensor $R_{\mu u}$ is governed by the characteristic substrate length $L_{\text{sub}} = c^2/a_0$ with dimensions $\mathrm{L^{-2}}$,

the mathematical structure satisfies rigorous dimensional homogeneity. Because this consistency is satisfied identically at the theoretical level, no modifications to the operational simulation code or node implementations are necessary. The framework stands on sound dimensional footing, bridging unfamiliar substrate terminology with canonical differential geometry and relativistic field theory.